

Introduction to Artificial Intelligence

Course Wrap-up

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Take-home Messages

- What is AI and how we define it in this course.
- Rational agents as a unifying framework for AI problems.
- Search as a core algorithmic technique for problem-solving.
- Logic as a formalism for knowledge representation and reasoning.
- Probabilistic models for reasoning under uncertainty.
- Planning as a way to connect reasoning and acting.

Class 01: What Is AI?

- Different views of AI: thinking vs acting, human-like vs rational.
- Turing Test and its limits as a definition of intelligence.
- Logic and cognitive inspiration as historical roots of AI.

Think like humans	Think rationally	Thinking
Act like humans	Act rationally	Acting
Like a human	Rationally	

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- Kind of agents:
 - simple reflex: condition-action rules;
 - model-based reflex: internal state to track unobservable aspects;
 - goal-based: explicit representation of goals to evaluate actions;
 - utility-based: explicit representation of preferences over states;
 - learning agents: improve performance through experience.

Take-home: Good AI design starts from a precise task formulation.

Class 03: Search I

- Problem formulation as a state-space search.
- Tree-search vs graph-search and the importance of avoiding repeated states.
- Complexity criteria: completeness, optimality, time, and space.
- Uninformed strategies and their trade-offs:

Algorithm	Completeness	Time complexity	Space complexity	Optimal?
Breadth-First	Yes	$O(n^d)$	$O(n^d)$	Depends on the cost function
Uniform Cost	Yes	Better than BFS	Better than BFS	Depends on the cost function
Depth-First	No	$O(n^m)$	$O(nm)$	No
Depth-Limited	No	$O(n^\ell)$	$O(n^\ell)$	No
Iterative Deepening	Yes	$O(n^d)$	$O(nd)$	Depends on the cost function

Take-home: Representation choices strongly affect search feasibility.

Running examples:

- search

Class 04: Search II

- Heuristics to guide search with domain knowledge.
- Greedy best-first vs A^* and the role of admissibility/consistency.
- Better heuristics = fewer explored states.

Running examples:

- heuristics

Class 05: Search III (Games)

- Adversarial search with game trees.
- Minimax for optimal play against an optimal opponent.
- Alpha-beta pruning to reduce explored branches.

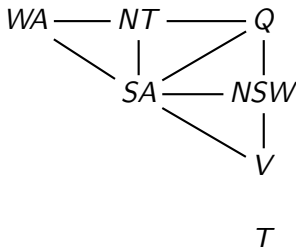
Take-home: Same core search ideas, but with competing objectives.

Running examples:

- minimax
- alpha-beta pruning

Class 06: Constraint Satisfaction Problems

- Variables, domains, and constraints as a factorized state model.
- Backtracking search + inference (consistency techniques) to prune the search space.
- Heuristics (Minimum Remaining Values, Degree heuristic) to choose variables/values.



Take-home: CSP structure can turn huge searches into manageable ones.

Class 10: MiniZinc Tutorial

- Modeling decision/optimization problems with constraints.
- Separation between model and data.
- Solving through different backends (CP, MIP, SAT) via FlatZinc.

Take-home: Declarative modeling is often faster than coding custom search.

Class 07: Logical Agents

- Knowledge-based agents: tell/ask cycle on a knowledge base.
- Propositional logic syntax and semantics, entailment ($\models$), and inference ($\vdash$).
- Inference algorithms: soundness and completeness.
- Resolution to check entailment by refutation.

Take-home: Logic gives explainable reasoning with formal guarantees.

Class 08: Inference in PL and Intro to FOL

- Resolution algorithm in propositional logic.
- Practical model-checking considerations.
- Motivation for first-order logic to express structure compactly.

Take-home: FOL increases expressiveness beyond propositional encodings.

Class 09: Unification and Inference in FOL

- Terms, predicates, quantifiers, and substitutions.
- Unification as the engine behind general FOL inference.
- Applying FOL to richer domains (e.g., Wumpus-style reasoning).

Take-home: General rules + unification enable compact, reusable knowledge.

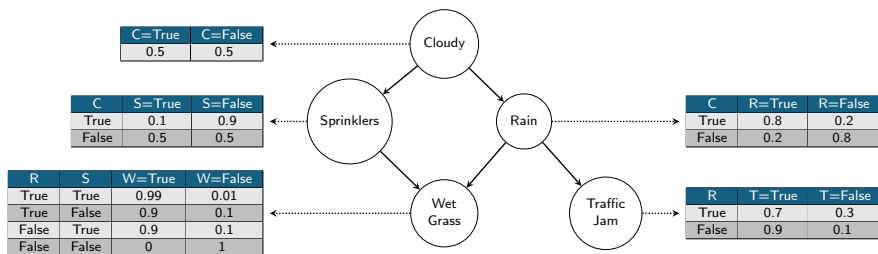
Class 11: Quantifying Uncertainty

- Why pure logic is insufficient under partial observability or non-determinism.
- Probability basics, conditional probability, and Bayes rule.
- Independence assumptions to simplify reasoning.

Take-home: Rational action under uncertainty needs probabilistic reasoning.

Class 12: Bayesian Networks

- Directed acyclic graphs encoding conditional dependencies.
- Compact factorization of joint distributions.
- Inference from evidence and causal/intuitive interpretation.



Take-home: Structure + probability gives scalable uncertain reasoning.

Class 13: Classical Planning and PDDL

- Planning with explicit action schemas: preconditions and effects.
- Domain/problem separation in PDDL.
- Plans as sequences that transform initial state into goal state.
- Closed-world assumption and its implications for reasoning.

Take-home: Symbolic planning connects logic representations and action.

Big Picture

- We moved from **agents and search** to **knowledge and logic**.
- Then from deterministic reasoning to **uncertainty and probabilistic models**.
- Finally, we returned to **acting**: planning with formal action models.

Core message: AI is about selecting actions rationally from models of the world.

Written Exam

Format:

- no book, no notes, no phone, no laptop, no internet.
- Part 1: multiple choice questions (MCQ) on all topics covering basic definitions, algorithms, and concepts.
- Part 2: open questions (OQ) requiring reasoning and application of concepts.

Grade

- if $< 60\%$ on MCQ, then fail regardless of OQ performance
- if $> 60\%$ on MCQ and pass on OQ, then max grade 2
- if $> 80\%$ on MCQ, then the grade is a combination of the weight of both parts ($\text{MCQ} \times 0.3 + \text{OQ} \times 0.7$)

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"That's all Folks!"